

Research on Adverse Drug Reactions of GLP-1RA Based on the FAERS Database

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ABSTRACT

OBJECTIVE To analyse and mine the adverse drug reaction (ADR) signals and off-label safety risks of six glucagon-like peptide-1 receptor agonists (GLP-1RA) using the U.S. FDA Adverse Event Reporting System (FAERS), thereby providing a reference for the rational clinical use of these medicines.

METHODS The online tool Open Vigil 2.1 was used to retrieve adverse drug event (ADE) reports submitted to the U.S. FAERS database from the first quarter of 2004 to the third quarter of 2023. The ADE were coded with the preferred terms (PT) of the Medical Dictionary for Regulatory Activities (MedDRA) and translated into Chinese. Signal mining was conducted by the proportional-imbalance method to screen and analyse the ADR signals.

RESULTS A total of 61 495, 24 553, 53 846, 1 985, 20 330 and 18 961 ADE reports were retrieved for exenatide, liraglutide, dulaglutide, lixisenatide, tirzepatide and semaglutide respectively. Women accounted for a higher proportion than men, and the principal reporting countries were the United States, the United Kingdom, France and Japan. The ADR signal with the highest number of cases was hyperglycaemia for exenatide and lixisenatide, nausea for liraglutide and semaglutide, and injection-site pain for tirzepatide and dulaglutide. The signal strength of liraglutide-induced pancreatitis was the highest among the six drugs (PRR=25.149). The gastrointestinal ADR with the lowest detection rates were epigastric pain and abdominal discomfort, and lixisenatide had the lowest gastrointestinal ADR detection rate.

CONCLUSION Using the FAERS database to retrieve the ADR signals of the six GLP-1RA since their marketing approval can provide a basis for clinical medication safety. To improve patients' medication adherence, particular attention should be paid to gastrointestinal adverse reactions, while new risk signals such as hyperglycaemia and weight gain should be monitored closely.

KEYWORDS glucagon-like peptide-1 receptor agonist; adverse drug reaction signals; FAERS database; pharmacovigilance

Introduction

In recent years, with socioeconomic development and changes in lifestyle, the prevalence of diabetes in China has continued to rise, increasing from less than 1% in 1980 to 10.6% in 2021^[1]. The International Diabetes Federation (IDF) predicts that by 2030 the prevalence of diabetes among people aged 20 to 79 in China will be approximately 11.8%^[2]. The disease burden of diabetes and its complications is growing ever heavier, and there is a pressing need for treatments that combine good efficacy with few adverse reactions. Glucagon-like peptide-1 (GLP-1) is an incretin that effectively enhances insulin sensitivity and promotes the proliferation of islet beta-cells^[3]. GLP-1 receptor agonists (GLP-1 RA) such as exenatide and liraglutide exert a marked hypoglycaemic effect by activating GLP-1; they act on GLP-1 receptors in the central and peripheral nervous systems, thereby controlling appetite and delaying gastric emptying to produce weight loss, and may also confer a degree of renal protection in patients with type 2 diabetes^[4]. In clinical trials, however, GLP-1 RA are prone to adverse drug reactions (ADR), such as gastrointestinal ADR^[5] (nausea, vomiting, diarrhoea and the like) and injection-site reactions^[6]. Using the online pharmacovigilance tool Open Vigil 2.1, this study mined the suspected adverse drug events (ADE) reported for the six GLP-1 RA above in the U.S. Food and Drug Administration (FDA) Adverse Event Reporting System (FAERS), screened and compared their principal ADE differences, and identified potential adverse-reaction signals so as to provide a reference for clinical medication safety.

1 Materials and Methods

1.1 Data source

All data in this study were drawn from the spontaneously reported and publicly released FAERS database. To facilitate the work, the pharmacovigilance analysis tool Open Vigil 2.1 (<https://openvigil.sourceforge.net>), which is updated in step with the FAERS database, was used for data mining to obtain ADE report data for the six GLP-1 RA from the first quarter of 2004 to the third quarter of 2023 (2004Q1–2023Q3).

1.2 Data extraction and processing

Open Vigil 2.1 offers efficient data-retrieval functions: data can be screened by entering the drug name, selecting the drug role and setting the ADE coding terms and retrieval period, so as to obtain basic information on the reported cases such as sex, age, outcome and reporting country. Each of the six GLP-1 RA was taken in turn as the study drug, and the ADE were coded in a standardised manner using the preferred terms (PT) of the Medical Dictionary for Regulatory Activities (MedDRA, version 26.1), compiled under the auspices of the International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH). All ADE data with report dates within 2004Q1–2023Q3 were obtained; the process is shown in Figure 1. Because the mined data contained multiple unrelated records for the same patient, only records for the target drug were retained for statistical analysis after screening. For example, 174 230 report records were retrieved for exenatide, of which the 61 495 ADE reports relating to exenatide were retained and the other-drug data for the same patients were removed.

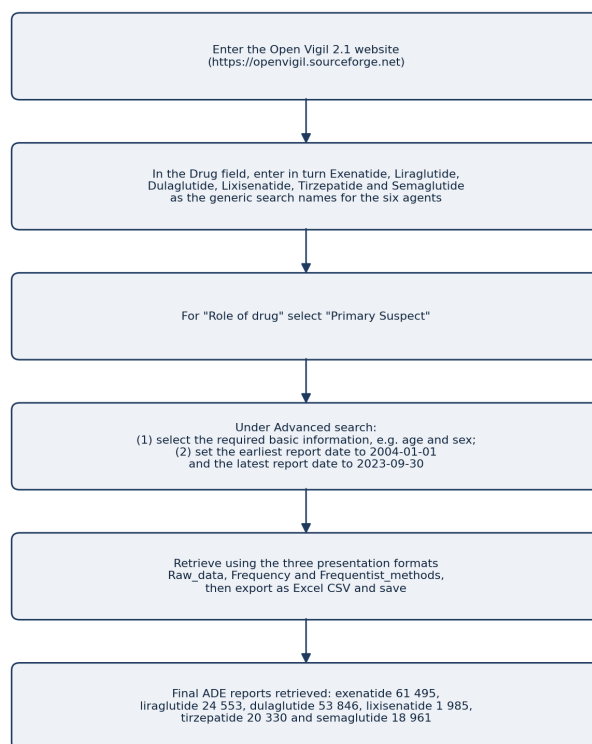


Figure 1 Data-retrieval flow chart based on Open Vigil 2.1

1.3 ADR signal mining

Open Vigil 2.1 mines ADE signals from the retrieved data using the proportional-imbalance method based on a four-fold table (Table 1), which comprises the proportional reporting ratio (PRR) and the reporting odds ratio (ROR). Once the ADE signal data for the six drugs had been obtained, the corresponding ADR signal data were screened out and analysed statistically. The larger the signal value, the greater the potential association between the drug and the target ADR. This study included only those ADR signals that were positive under both the ROR and the PRR method. The formulas and signal-detection criteria for the two algorithms are shown in Table 2.

Table 1 Four-fold table

Item	Target ADE	Other ADE	Total
Target drug	a	b	a + b
Other drugs	c	d	c + d
Total	a + c	b + d	N = a + b + c + d

Note: a – number of reports of the target ADE caused by the target drug; b – number of reports of other ADE caused by the target drug; c – number of reports of the target ADE caused by other drugs in the database; d – number of reports of other ADE caused by other drugs in the database; N – total number of ADE reports for all drugs in the database.

Table 2 Signal-detection methods and signal-threshold standards

Method	Formula	Positive-signal criterion
ROR method	$ROR = (a/c)/(b/d) = ad/bc$; $95\%CI = \exp[\ln(ROR) \pm 1.96 \cdot SE]$; $SE(\ln ROR) = \sqrt{1/a + 1/b + 1/c + 1/d}$	Lower limit of 95%CI > 1 and a ≥ 3 indicate one positive ADR signal (P < 0.05)
PRR method	$PRR = [a/(a+b)] / [c/(c+d)]$; $95\%CI = \exp[\ln(PRR) \pm 1.96 \cdot SE]$; $SE(\ln PRR) = \sqrt{1/a - 1/(a+b) + 1/c - 1/(c+d)}$	PRR > 1, chi-square ≥ 4 and a ≥ 3 indicate one positive ADR signal (P < 0.05)

2 Results

2.1 Basic information on the drug ADE

Using Open Vigil 2.1, ADE reports with exenatide, liraglutide, dulaglutide, lixisenatide, tirzepatide and semaglutide as the primary suspect were retrieved and screened, numbering 61 495, 24 553, 53 846, 1 985, 20 330 and 18 961 respectively. The basic information and demographic characteristics of the reports, after collation, are shown in Table 3. This study focused on the ADE reported for injectable administration; because the ADR signals for oral administration were weak, they were not analysed. The sex distribution of patients who developed ADR after subcutaneous injection of the six drugs was imbalanced, with women outnumbering men in every case; for tirzepatide the number of women with ADR was nearly four times that of men, while for the remaining drugs it was around twofold.

As regards age, patients aged 18 to 60 and those over 60 accounted for comparable proportions, tirzepatide alone excepted (for which the former was about four times the latter). The ADE outcomes following injection of these GLP-1 RA included hospitalisation or prolonged hospitalisation, disability, life-threatening events and even death; most patients with an ADE presented chiefly with hospitalisation or prolonged hospitalisation and with disability, semaglutide accounting for the highest proportion of these outcomes, whereas liraglutide accounted for the largest proportion of life-threatening and fatal outcomes. The overall outcome profile of tirzepatide was better than that of the other five drugs.

Since exenatide came onto the market in 2004, the ADE reports for 2004–2008 accounted for about half of its total and declined gradually thereafter; liraglutide was launched in 2009, its ADE reports accounting for nearly half during 2009–2013 and then decreasing; dulaglutide, lixisenatide, tirzepatide and semaglutide were approved by the FDA for marketing in the United States in 2014, 2012, 2022 and 2017 respectively, and their ADE report numbers have risen steadily since launch. Tirzepatide was the most recently marketed GLP-1 RA (2022), yet during 2019–2023 it recorded the highest number of ADE reports among the six drugs. In terms of geographical distribution, the United States returned the largest number of ADE reports, which may relate to the fact that the FAERS database is established and operated in the United States, so that local healthcare professionals and patients find it more convenient to submit reports, the proportion of reports from other countries being comparatively small.

Table 3 Basic information of ADE reports related to the six GLP-1 RA / %

Item	Exenatide (n=61 495)	Liraglutide (n=24 553)	Dulaglutide (n=53 846)	Lixisenatide (n=1 985)	Tirzepatide (n=20 330)	Semaglutide (n=18 961)
Sex						
Male	38.40	32.54	37.76	36.78	17.37	34.35
Female	58.63	61.75	51.32	46.35	67.87	60.88
Not reported	2.97	5.71	10.92	16.88	14.76	4.77
Age						
< 18 years	0.04	0.22	0.05	0.00	0.01	0.09
18–60 years	29.82	32.44	12.74	20.15	28.14	21.92
> 60 years	31.40	28.24	15.59	37.58	7.74	25.84
Not reported	38.74	39.11	71.61	42.27	64.11	52.15

Item	Exenatide (n=61 495)	Liraglutide (n=24 553)	Dulaglutide (n=53 846)	Lixisenatide (n=1 985)	Tirzepatide (n=20 330)	Semaglutide (n=18 961)
Outcome						
Hospitalisation / prolonged stay	10.32	13.47	6.16	6.35	2.86	13.21
Disability	0.32	0.55	0.29	0.81	0.10	1.15
Life-threatening	0.81	1.46	0.35	0.40	0.17	1.27
Death	2.08	2.62	1.32	0.65	0.24	1.05
Other / not reported	86.46	81.90	91.87	91.79	96.63	83.32
Year						
2004–2008	48.40	–	–	–	–	–
2009–2013	13.87	46.03	–	0.05	–	–
2014–2018	18.86	36.01	25.28	16.47	–	4.10
2019–2023	18.87	17.97	74.72	83.48	100.00	95.90
Reporting country						
United States	94.70	84.02	95.13	91.44	99.09	86.29

Note: “–” indicates no data. Only the United States row, which is reported for all six drugs, is tabulated here. The source lists only the top five reporting countries for each drug; the remaining reported countries and their percentages are: United Kingdom (1.09, 2.31, 0.40, 1.91), Germany (0.65, 0.43), China (0.39), Australia (0.31), Brazil (0.59, 1.06), France (1.22, 0.44, 0.07, 0.89), Japan (1.21, 1.30, 1.31, 0.36, 1.66), Israel (1.16), Russia (0.60), United Arab Emirates (0.29), Saudi Arabia (0.05) and Canada (1.82). In the original the per-drug attribution of these secondary figures is presented sparsely; readers should consult the source table for the exact drug-by-country alignment.

2.2 ADR signal detection results based on PT terms

The 15 ADR signals ranked highest by number of cases for the six GLP-1 RA are shown in Table 4, and the 10 ADR signals ranked highest by signal strength are shown in Table 5. For **exenatide**, the ADR signal ranked first by number of cases was hyperglycaemia, with 14 454 reports, and it does not appear in the prescribing information; the signal ranked first by strength was early satiety (PRR=191.845, lower limit of the 95%CI of ROR=170.563). For **liraglutide**, the signal ranked first by number of cases was nausea, with 4 676 reports, and one signal (hyperglycaemia) does not appear in the prescribing information; the signal ranked first by strength was thyroid C-cell hyperplasia (PRR=765.760, lower limit of the 95%CI of ROR=127.962). For **dulaglutide**, the signal ranked first by number of cases was injection-site pain, with 7 163 reports, and two signals (hyperglycaemia and increased glycated haemoglobin) do not appear in the prescribing information; the signal ranked first by strength was diabetic amyotrophy (PRR=68.721, lower limit of the 95%CI of ROR=22.408).

For **lixisenatide**, the signal ranked first by number of cases was hyperglycaemia, with 286 reports, and six signals (hyperglycaemia, visual impairment, increased glycated haemoglobin, weight gain, hearing loss and cataract) do not appear in the prescribing information; the signal ranked first by strength was colour blindness (PRR=47.046, lower limit of the 95%CI of ROR=17.611). For **tirzepatide**, the signal ranked first by number of cases was injection-site pain, with 2 784 reports, and one signal (hyperglycaemia) does not appear in the prescribing information; the signal ranked first by strength was injection-site coldness (PRR=70.987, lower limit of the 95%CI of ROR=49.796). For **semaglutide**, the signal ranked first by number of cases was nausea, with 3 433 reports, and two signals (hyperglycaemia and weight gain) do not appear in the prescribing information; the signal ranked first by strength was starvation ketoacidosis (PRR=138.567, lower limit of the 95%CI of ROR=61.314).

Table 4 Signal-mining results for the top 15 ADR by incidence frequency of the six GLP-1 RA

No.	Exenatide	n	Liraglutide	n	Dulaglutide	n	Lixisenatide	n	Tirzepatide	n	Semaglutide	n
1	Hyperglycaemia*	14 454	Nausea	4 676	Injection site pain	7 163	Hyperglycaemia*	286	Injection site pain	2 784	Nausea	3 433
2	Nausea	10 402	Hyperglycaemia*	1 946	Nausea	6 366	Visual impairment*	97	Nausea	2 490	Vomiting	2 087
3	Injection site haemorrhage	5 741	Vomiting	1 931	Hyperglycaemia*	6 434	Nausea	72	Injection site haemorrhage	1 277	Diarrhoea	1 614
4	Injection site pain	5 101	Diarrhoea	1 826	Injection site haemorrhage	2 405	Blood glucose abnormal	46	Diarrhoea	1 078	Constipation	930
5	Vomiting	3 846	Pancreatitis	1 568	Glycated haemoglobin increased*	1 710	Glycated haemoglobin increased*	37	Vomiting	940	Headache	921
6	Pancreatitis	2 623	Epigastric pain	873	Epigastric pain	1 411	Diabetes poorly controlled	37	Injection site erythema	887	Hyperglycaemia*	760
7	Injection site mass	2 615	Constipation	806	Injection site bruising	1 379	Weight gain*	30	Constipation	663	Fatigue	699
8	Injection site bruising	2 456	Abdominal discomfort	620	Abdominal discomfort	1 162	Pancreatitis	28	Injection site bruising	578	Epigastric pain	691
9	Diarrhoea	2 431	Abdominal pain	622	Injection site erythema	1 133	Blood glucose fluctuation	23	Hyperglycaemia*	495	Pancreatitis	513
10	Dizziness	2 254	Pancreatic carcinoma	572	Pancreatitis	1 106	Hearing loss*	22	Injection site pruritus	493	Abdominal distension	502
11	Injection site extravasation	2 243	Asthenia	537	Injection site mass	978	Injection site haemorrhage	19	Eructation	419	Abdominal pain	500
12	Injection site nodule	2 010	Acute cholecystitis	497	Malaise	946	Injection site pain	13	Epigastric pain	382	Abdominal discomfort	447

No.	Exenatide	n	Liraglutide	n	Dulaglutide	n	Lixisenatide	n	Tirzepatide	n	Semaglutide	n
13	Headache	1 750	Dyspepsia	473	Abdominal pain	872	Injection site swelling	13	Malaise	372	Eructation	421
14	Paraesthesia	1 615	Injection site erythema	424	Injection site pruritus	871	Injection site bruising	12	Dyspepsia	330	Dehydration	421
15	Malaise	1 548	Injection site pain	414	Constipation	864	Cataract*	11	Abdominal discomfort	319	Weight gain*	350

Note: * – ADR not listed in the official prescribing information (because the prescribing information states that the specific manifestations of gastrointestinal reactions vary between individuals, gastrointestinal ADR were not treated as unlisted). The prescribing information is taken from exenatide (BYDUREON), liraglutide (VICTOZA), dulaglutide (TRULICITY), lixisenatide (ADLYXIN), tirzepatide (MOUNJARO) and semaglutide (OZEMPIC).

Table 5 Signal-mining results for the top 10 ADR by signal strength of the six GLP-1 RA

(a) *Exenatide, liraglutide and dulaglutide*

No.	Exenatide			Liraglutide			Dulaglutide		
	PT	PRR	ROR 95%CI lower	PT	PRR	ROR 95%CI lower	PT	PRR	ROR 95%CI lower
1	Early satiety	191.845	170.563	Thyroid C-cell hyperplasia	765.760	127.962	Diabetic amyotrophy	68.721	22.408
2	Injection site nodule	98.939	96.820	Blood calcium increased	113.447	63.604	Injection site coldness	32.605	23.414
3	Injection site extravasation	74.323	73.220	Medullary thyroid carcinoma	72.285	42.845	Medullary thyroid carcinoma	20.820	11.225
4	Localised urticaria	62.344	28.326	Thyroid cancer	47.489	23.017	Intraductal papillary mucinous neoplasm	15.228	6.660
5	Food regurgitation	61.083	36.154	Follicular thyroid cancer	38.098	15.358	Injection site pain	13.835	6.449
6	Injection site indentation	44.445	37.533	Eructation	32.316	30.144	Injection site haemorrhage	12.855	12.844
7	Injection site cyst	33.302	21.247	Obstructive cholelithiasis	30.478	11.115	Thyroid mass	12.805	10.190
8	Hyperglycaemia	31.418	40.307	Intraductal papillary mucinous neoplasm	28.680	11.652	Obstructive pancreatitis	11.868	6.472
9	Injection site deformation	30.685	13.886	Pancreatic carcinoma	26.768	18.323	Injection site mass	11.472	10.922
10	Injection site mass	29.325	29.278	Lipase increased	25.849	22.410	Projectile vomiting	11.002	8.498

(b) *Lixisenatide, tirzepatide and semaglutide*

No.	Lixisenatide			Tirzepatide			Semaglutide		
	PT	PRR	ROR 95%CI lower	PT	PRR	ROR 95%CI lower	PT	PRR	ROR 95%CI lower
1	Colour blindness	47.046	17.611	Injection site coldness	70.987	49.796	Starvation ketoacidosis	138.567	61.314
2	Hypoglycaemic coma	25.868	9.699	Injection site pain	53.215	46.598	Medullary thyroid carcinoma	105.931	64.428
3	Visual impairment	9.651	8.215	Injection site paraesthesia	30.003	21.174	Obstructive pancreatitis	73.323	47.235
4	Lipase increased	8.435	3.513	Eructation	29.739	27.441	Allodynia	67.921	47.786
5	Gastric emptying delayed	7.132	2.677	Injection site laceration	19.082	10.471	Injection site electric-discharge sensation	43.749	34.641
6	Acute pancreatitis	5.684	3.066	Injection site haemorrhage	17.740	17.798	Eructation	31.953	29.476
7	Hearing loss	5.565	3.690	Thirst decreased	16.048	7.575	Injection site extravasation	30.050	27.431
8	Pancreatitis	5.380	3.694	Gastric emptying delayed	13.198	10.578	Proliferative retinopathy	25.871	9.526
9	Injection site extravasation	4.072	1.696	Allodynia	11.560	5.472	Diabetic retinopathy	22.553	17.324
10	Injection site rash	3.735	1.947	Injection site urticaria	11.332	10.034	Injection site pain	22.526	8.315

Note: PRR – proportional reporting ratio; the third column of each pair is the lower limit of the 95% confidence interval of the ROR. Prescribing information sources as for Table 4.

2.3 Detection rates of the main gastrointestinal ADR signals

Gastrointestinal adverse reactions are the most common ADR of the GLP-1 RA class. For all six drugs the gastrointestinal ADR signal with the highest detection rate was nausea, liraglutide having the largest proportion (8.74%), followed by semaglutide (6.70%), and lixisenatide the smallest (1.85%). In terms of signal strength, the nausea signal caused by liraglutide was the strongest (PRR=5.783, lower limit of the 95%CI of ROR=6.704). Liraglutide also had the highest proportion of pancreatitis among the six drugs (2.93%); the rates of vomiting (liraglutide 3.61%, semaglutide 4.07%) and diarrhoea (3.41% and 3.15%) for liraglutide and semaglutide were higher than for the other four drugs. Overall, the gastrointestinal ADR signals with the lowest detection rates were epigastric pain and abdominal discomfort, and compared with the other GLP-1 RA, lixisenatide had a lower gastrointestinal ADR detection rate. The results are shown in Figure 2.

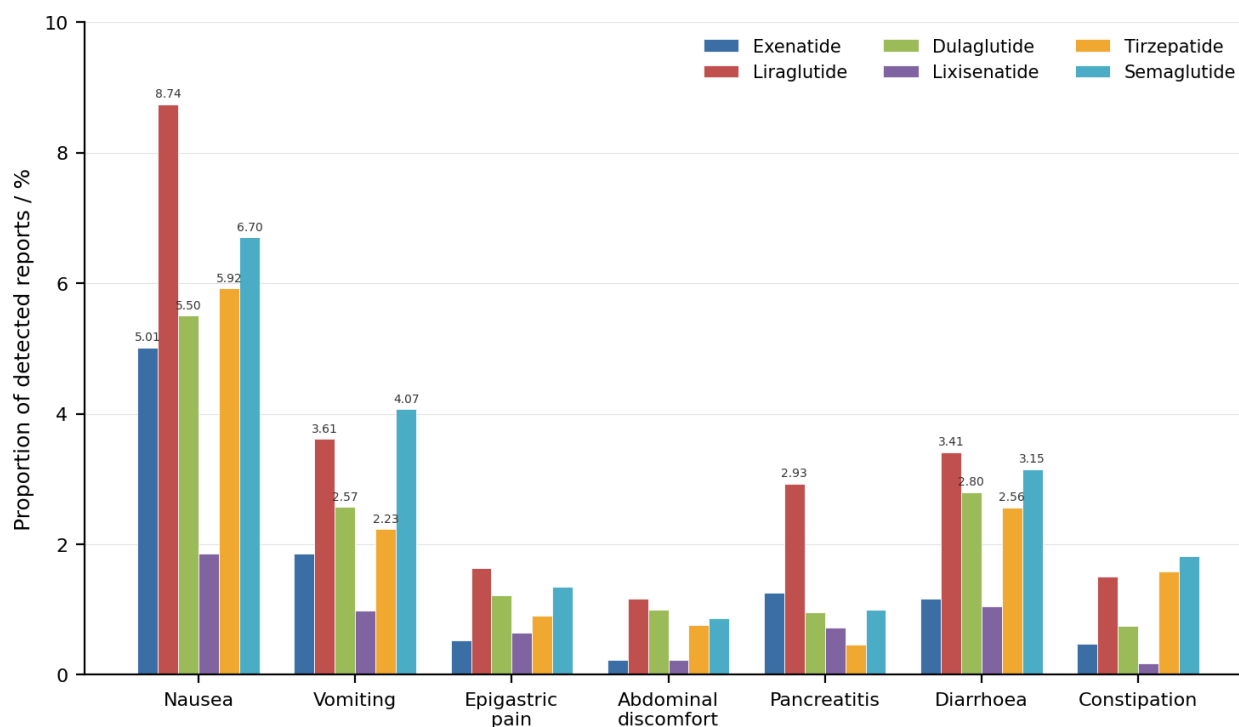


Figure 2 Detection rate of the major gastrointestinal ADR signals in the six GLP-1 RA

3 Discussion

3.1 Drug background and ADR already known in the prescribing information

As regards the sex distribution in the basic demographic profile, for all six GLP-1 RA the proportion of women with ADR was higher than that of men, which may relate to differences in drug use between the sexes or to limitations in the clinical observational data. On the one hand, many women use GLP-1 RA as weight-loss drugs, and a high body-fat percentage may promote accumulation of the drug in adipose tissue and thereby cause ADR^[7]. On the other hand, women are more sensitive in their metabolic response to certain drugs^[8] and may be more prone to ADR. Moreover, data bias and reporting bias may affect the sex ratio, since in everyday medication use women are more inclined to report ADR actively. Future studies could explore this sex difference further in order to explain and control the risk of ADR more effectively.

The principal ADR signals of the six GLP-1 RA screened in this study were gastrointestinal reactions, manifesting as nausea, vomiting, abdominal pain, constipation and dyspepsia, with liraglutide, tirzepatide and semaglutide ranking highest for the detection rate of gastrointestinal-reaction signals. Earlier work has shown^[9] that these ADR are seen mostly at the start of treatment and are transient; as treatment continues, the body's tolerance increases and the symptoms ease gradually. The clinical-trial results recorded in the prescribing information^[10] show that gastrointestinal reactions occurred in 23.2% and 70.1% of patients in the placebo and treatment groups respectively for exenatide; 14.0% and 53.0% for liraglutide; 25.7% and 47.0% for dulaglutide; 14.0% and 43.0% for lixisenatide; 20.0% and 48.0% for tirzepatide; and 11.8% and 34.3% for semaglutide. These results indicate that the incidence of gastrointestinal ADR has fallen steadily with successive generations of the drugs, the more recently marketed GLP-1 RA having a lower incidence.

This study also found a relatively strong association between GLP-1 RA and pancreatitis. The pancreatitis ADR signals for all six GLP-1 RA were relatively strong, the order of risk being liraglutide > exenatide > semaglutide > dulaglutide > lixisenatide > tirzepatide, which suggests that clinicians using GLP-1 RA should pay particular attention to patients with sudden epigastric pain during treatment so as to ensure medication safety. In a historical cohort study of adults with type 2 diabetes^[11], no increase in the incidence of pancreatic cancer was observed within seven years of GLP-1 RA treatment, although the investigators recommended that the risk be monitored for seven years after treatment. Drawing on the openFDA database, Dong Shichao et al.^[12] found that liraglutide and four other drugs all produced moderate-to-strong (acute) pancreatitis ADR signals, with a relatively high risk of occurrence. At present the effect of GLP-1 RA on the human pancreas remains a matter of debate and requires further analysis through high-quality clinical

trials.

3.2 Risk signals not listed in the prescribing information

This study analysed the ADR signals of the six GLP-1 RA and identified suspected high-risk signals, ranking highly by both number of cases and signal strength, that are not mentioned in the prescribing information and warrant greater attention during clinical use.

3.2.1 Hyperglycaemia

Hypoglycaemia is a common ADR of glucose-lowering drugs; sulfonylureas, which act for a long time and take effect quickly, very readily induce hypoglycaemic symptoms. GLP-1 RA, by contrast, promote insulin secretion in a glucose-concentration-dependent manner, inhibit glucagon secretion and delay gastric emptying, reducing food intake by suppressing appetite and so lowering blood glucose^[13]. Used alone, GLP-1 RA seldom cause hypoglycaemia; this arises mostly from co-administration with other fast-acting glucose-lowering drugs, which raises the risk. It is worth noting that, among the adverse signals mined in this study, all six GLP-1 RA showed raised blood glucose and raised glycated haemoglobin (HbA1c) in very large numbers, and these spontaneously reported data all come from the real world. The causes may be analysed as follows. (1) The dose of GLP-1 RA may have been insufficient, leaving blood glucose persistently high. The FAERS data used here derive from a spontaneous-reporting system and therefore cover a very wide range of report types, including reports of inadequate efficacy, so ADR signals due to insufficient dosage may arise. (2) When diabetic patients combine GLP-1 RA with an excess of other glucose-lowering drugs, blood glucose falls rapidly; this change triggers the body's self-protective response, causing a sharp rise in glucose-raising hormones and so inducing reactive hyperglycaemia, also known as the Somogyi effect. (3) It may relate to particular alleles: a prospective cohort study^[14] found that subjects carrying the GLP1R rs3765467 GG genotype showed a markedly greater reduction in HbA1c than those with the GA+AA genotype $[(1.7\pm2.4)\% \text{ vs } (0.8\pm1.8)\%; P=0.002]$, and that GLP1R rs3765467 polymorphism is associated with the treatment response to GLP-1 RA in Chinese patients with type 2 diabetes. Causes (1) and (2) are linked to poor lifestyle habits and irregular medication; it is recommended that, when prescribing and diagnosing, physicians stress to patients the importance of regular medication, correct diet and appropriate exercise. Cause (3) is closely tied to patients' genetic characteristics, and it is conjectured that the hyperglycaemia ADR may relate to the smaller glucose-lowering benefit in patients carrying the GA+AA genotype.

3.2.2 Weight gain

The 2023 American Diabetes Association (ADA) guidelines^[15] rank glucose-lowering drugs by their weight-loss effect, classing semaglutide as a drug with "very good weight-loss effect" and dulaglutide and liraglutide as drugs with "good weight-loss effect". Through a systematic review and meta-analysis (seven trials, 6 609 subjects), Karagiannis et al.^[16] concluded that the dual gastric inhibitory polypeptide (GIP)/GLP-1 receptor agonist tirzepatide is more effective than the GLP-1 RA in reducing body weight. The data-mining results of this study, however, show that both lixisenatide and semaglutide produced a weight-gain ADR signal (ranked 7th and 15th respectively by number of cases), which differs from the previously observed tendency of GLP-1 RA to cause weight loss through ADR such as diarrhoea and dehydration. The main causes are analysed as: (1) a feeling of hunger after medication, so that patients eat more and gain weight; and (2) excessive injected dosage, so that the body synthesises more glycogen and protein and thereby gains weight. Clinically, attention should be paid to the potential risk of obesity caused by overmedication in non-obese patients with type 2 diabetes.

3.2.3 Visual impairment

According to the lixisenatide prescribing information, the drug is not stated to cause serious visual impairment. This study, however, found that lixisenatide may induce visual impairment (ranked 2nd by number of cases) and cataract (ranked 15th), and may cause colour blindness (ranked 1st by signal strength); semaglutide likewise carries a potential risk of inducing retinopathy. The probability and severity of ocular ADR caused by GLP-1 RA require careful study and attention. Such ADR may relate to the fall in blood glucose after medication, since a drop in blood glucose can alter the osmotic pressure of the lens and affect its accommodative capacity, leading to changes in vision, blurred vision and similar symptoms. The following causes should be excluded: (1) patients taking too small a dose, or not injecting the glucose-lowering drug on time or in the correct amount, so that blood glucose stays elevated and affects the retina; and (2) long-standing poor glycaemic control, which itself can cause cataract, retinopathy, glaucoma and other eye diseases that lead to visual impairment.

3.3 Analysis of the main gastrointestinal ADR signals

The data-mining results show that, among the gastrointestinal ADR caused by GLP-1 RA, nausea had the highest number of cases; the incidence of nausea with liraglutide was higher than with the other drugs, the order by signal strength being liraglutide > semaglutide > exenatide > tirzepatide > dulaglutide > lixisenatide. An ADR/ADE analysis report covering 28 GLP-1 RA cases^[17] noted that, among 12 digestive-system events, the incidence with liraglutide was twice that with exenatide, broadly consistent with the present results. To date no clinical study has confirmed a causal relationship between GLP-1 RA and pancreatitis; from the real-world reports collected here, pancreatitis appeared among the possible ADR signals of all six GLP-1 RA, and the pancreatic adenocarcinoma signal for liraglutide (PRR=26.768) and the obstructive-pancreatitis signal for dulaglutide (PRR=11.868) both suggest that GLP-1 RA can increase the risk of pancreatitis. In a real-world disproportionality study based on the FAERS database, Liu et al.^[18] demonstrated the association between different GLP-1 RA and gastrointestinal ADR, finding that liraglutide carried the greatest risk of epigastric pain (ROR=4.63; 95%CI=4.12–5.21) and pancreatitis (ROR=32.67; 95%CI=29.44–36.25), again consistent with the present findings. A health-technology assessment incorporating 12 systematic reviews and meta-analyses^[19] showed that semaglutide can significantly lower HbA1c, fasting blood glucose and systolic blood pressure and reduce body weight without increasing the incidence of acute pancreatitis. A real-world cohort study by Faillie et al.^[20] (25 617 subjects) showed that the incidence of intestinal obstruction associated with GLP-1 RA treatment increased 3.5-fold, and that the risk of chronic intestinal obstruction accumulated gradually over time, peaking after 1.6 years of treatment. Analysing high-quality randomised-controlled-trial data on the constipation incidence of four GLP-1 RA against treatment duration, Lu et al.^[21] found that constipation incidence was positively correlated with treatment duration ($r^2 = 0.8\text{--}0.9$), suggesting that long-term use of GLP-1 RA may increase the incidence of chronic intestinal adverse events.

3.4 Limitations

Because the raw data in the FAERS database are complex and highly heterogeneous, and the reported drug names are not compulsorily standardised, retrieving and cleaning the data is difficult. This study used Open Vigil 2.1, an online pharmacovigilance data-analysis platform built by overseas scholars, which allows retrieval directly by drug name without downloading the complete raw FAERS data and performing onerous cleaning and preprocessing, thereby avoiding the handling of large volumes of useless data; this feature is convenient for researchers who wish to use the FAERS database but do not yet have large-scale data-processing capacity. The platform nonetheless has limitations: some original reports submitted to FAERS may be impossible to retrieve because the drug name was misspelt or the drug was not placed in the correct category, producing unavoidable bias. Recently, Chinese scholars using STATA software compared the FAERS raw data with the Open Vigil 2.1 platform data^[22] and found that the raw-data retention of Open Vigil version 2.1 was 74.1%. In addition, some original reports lacked baseline information such as sex and age, which may bias the demographic analysis. The main purpose of this study was to analyse drug-specific ADR in order to identify potential drug-related risk signals, rather than to assess broader ADE (such as operational errors or problems with the route of administration). The ADE reports in the FAERS database do not clearly distinguish drug relatedness from background event factors (such as patient history or medical errors), which introduces a degree of bias into the ADR analysis; however, since the ADR signals of the six drugs studied are mostly induced after medication and are largely related to the drugs themselves, this does not affect the overall trend of the results. Because other ADE may exert a potential influence on signal detection, future studies could assess the degree of such interference by combining detailed patient data with multivariate analysis.

4 Conclusion

Based on the FAERS database, this study systematically analysed the principal and potential ADR signals of six GLP-1 RA and further examined their adverse-reaction profile and safety characteristics. On the basis of the findings, it is recommended that, when using GLP-1 RA, clinicians pay particular attention to gastrointestinal adverse reactions (such as nausea and epigastric pain) and remain alert to the newly identified risk signals (such as hyperglycaemia and weight gain), adjusting the treatment regimen promptly according to the specific circumstances. For patients with marked gastrointestinal reactions, for example, the dose may need to be adjusted or the drug changed; for patients with raised blood glucose or weight gain, combination with other drugs or adjustment of the GLP-1 RA regimen may be considered. In view of possible sex differences in drug use (this study suggesting that female patients are more prone to adverse reactions), individualised treatment strategies should also be strengthened. In sum, this study provides important scientific evidence and a medication reference for the rational clinical use of GLP-1 RA, with the aim of helping physicians optimise treatment strategies, attend to adverse reactions and improve patients' treatment adherence and

quality of life.

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